Notes: Board & Meyer-ter-Vehn (Econometrica, 2021)

Learning Dynamics in Social Networks

Contents

1 Big picture

- **Question.** How does network structure determine Bayesian learning and diffusion of an innovation when agents observe neighbors' adoption decisions?
- **Core setting.** Agents enter over time and decide whether to inspect a product. Inspection is costly but reveals product quality perfectly. Agents observe whether neighbors have adopted, not their entry times or inspection choices.
- **Main object.** The paper studies *social learning curves*: the probability that an agent has seen at least one neighbor adopt by the time she moves.
- **Methodological contribution.** Instead of solving Bayesian updating on an exact finite network, the paper studies large random networks whose local structure is summarized by a small number of state variables and ordinary differential equations.
- **Main directed-network result.** In directed locally-tree-like networks, additional direct and indirect links improve social learning and welfare, under a bounded hazard rate condition on inspection costs.
- **Directed versus undirected result.** Directed networks generate better learning than comparable undirected networks because backward links create self-reflection: a neighbor who observes the agent becomes more pessimistic when the agent has not yet adopted.
- **Clustering result.** Cliques and triangles reduce learning relative to the same number of independent bilateral links. Clustering correlates neighbors' decisions exactly when independent experimentation would be most valuable.
- **Correlation-neglect result.** If agents ignore correlation among their information sources, they overestimate how likely they are to see an adoption and become too pessimistic after seeing none. This reduces learning and welfare.
- **Dense-network result.** Sparse networks can aggregate information perfectly as average degree grows, but adding many clustered links can lower welfare and prevent full information aggregation.
- **Economic takeaway.** Connectedness helps when it adds independent sources of information. It hurts when it creates redundant, correlated, or self-reflective information paths.

2 Model primitives and measurement

2.1 Agents, product quality, and timing

- There is a finite set of agents connected by an exogenous directed network $G \subseteq I \times I$.
- If i observes j, write $i \to j$ and call j a neighbor of i.
- Product quality is binary, $q \in \{H, L\}$, with prior $\pi_0 = \Pr(q = H)$.
- Each agent enters at a random time and decides whether to inspect the product.
- Inspection cost κ_i is drawn from distribution F.
- Inspection reveals quality perfectly; high-quality products are adopted, low-quality products are not adopted.
- Agents observe neighbors' adoption decisions, but not their entry times or inspection choices.

Interpretation: The model cleanly separates *social information*, which affects whether an agent inspects, from *private information*, which is perfect after inspection and determines adoption.

2.2 Posterior after no adoption

If a neighbor has adopted, the agent infers $q = H$. If no neighbor has adopted, and the probability of observing a neighbor's adoption under $q = H$ is y, Bayes' rule gives

$$\pi(y) = \frac{(1-y)\pi_0}{(1-y)\pi_0 + (1-\pi_0)}. \tag{1}$$

Interpretation: Seeing no adoption is bad news. The larger y is, the more informative non-adoption becomes, and the lower the posterior $\pi(y)$.

2.3 Adoption and social learning curves

Let x_i be the probability that agent i adopts the high-quality product by time t. Let y_i be the probability that at least one of i's neighbors has adopted by that time. The realized adoption rate follows

$$\dot{x}_{i,G,\xi} = 1 - (1 - y_{i,G,\xi})\,(1 - F(\pi(y_{i,\xi_i}))) = \phi(y_{i,G,\xi}, y_{i,\xi_i}). \tag{2}$$

Taking expectations gives the interim adoption curve

$$\dot{x}_{i,\xi_i} = 1 - (1 - y_{i,\xi_i})\,(1 - F(\pi(y_{i,\xi_i}))) = \Phi(y_{i,\xi_i}). \tag{3}$$

Interpretation: y summarizes social learning. x is behavior. The paper compares networks by comparing their induced y curves.

2.4 Bounded hazard rate condition

The key monotonicity condition is

$$\frac{f(\kappa)}{1-F(\kappa)} \leq \frac{1}{\kappa(1-\kappa)}, \qquad \kappa \in [0, \pi_0]. \tag{4}$$

Interpretation: Better social information has two effects: observing adoption encourages adoption, but observing no adoption becomes worse news. The bounded hazard rate condition ensures the first force dominates in expectation, so more social information increases adoption.

2.5 Welfare

Because low-quality products are never adopted, welfare costs come from two sources:

- failing to adopt a high-quality product;
- paying inspection costs for a low-quality product.

Interpretation: A higher social learning curve Blackwell-improves information and raises welfare.

3 Models and empirical-style specifications

3.1 Directed pair and directed chain

In a directed pair $i \to j$, j has no social information, while i learns from j. If x_j is j's adoption curve, i's adoption rate is

$$\dot{x}_i = 1 - (1 - x_j)(1 - F(\pi(x_j))) = \Phi(x_j). \tag{5}$$

For an infinite directed chain, symmetry gives the one-dimensional ODE

$$\dot{x} = \Phi(x). \tag{6}$$

Interpretation: The chain already shows the paper's key simplification: Bayesian learning dynamics can be summarized by a low-dimensional ODE.

3.2 General directed random networks

In a directed random network with type-specific degree vector $d = (d_\theta)$, the adoption curve solves

$$\dot{x}_\theta = E\left[\Phi\left(1 - \prod_{\theta'}(1 - x_{\theta'})^{D_{\theta,\theta'}}\right)\right]. \tag{7}$$

For a regular single-type network with degree d,

$$\dot{x} = \Phi\left(1 - (1 - x)^d\right). \tag{8}$$

Interpretation: Large directed random networks are locally tree-like, so neighbors' adoption decisions are approximately independent. This delivers tractable diffusion curves.

3.3 Undirected random networks

In undirected networks, the degree distribution of a typical neighbor is weighted by the friendship paradox:

$$\Pr(D' = d) = \frac{d}{E[D]} \Pr(D = d). \tag{9}$$

The undirected adoption process solves

$$\dot{\bar{x}} = E\left[\phi\left(1 - (1 - \bar{x})^{D'-1}, 1 - (1 - \bar{x})^{D'}\right)\right]. \tag{10}$$

Interpretation: Undirected links create a self-reflection problem. A neighbor's failure to adopt is partly caused by observing the agent's own non-adoption, which makes the neighbor too pessimistic from the agent's perspective.

3.4 Clustering and triangles

When agents are embedded in triangles, define $\hat{x}$ as the first-adopter probability inside a triangle. The clustered-network ODE is

$$\dot{\hat{x}} = E\left[\phi\left(1 - (1 - \hat{x})^{2(D'-1)}, 1 - (1 - \hat{x})^{2D'}\right)\right]. \tag{11}$$

Interpretation: Clustering correlates neighbors' actions. If one neighbor fails to adopt, other neighbors are more likely to fail too. This reduces the probability that the agent sees at least one adoption.

3.5 Correlation neglect

Agents in triangular networks may believe their information sources are independent. In that case, subjective beliefs use the bilateral-link curve $\bar{x}$ while objective adoption follows

$$\dot{\check{x}} = E\left[\phi\left(1 - (1 - \check{x})^{2(D'-1)}, 1 - (1 - \bar{x}^*)^{2D'}\right)\right]. \tag{12}$$

Interpretation: Correlation neglect makes agents overestimate how informative no adoption should be, so they become excessively pessimistic and inspect less.

4 Identification and theoretical design

4.1 What is being compared

The paper compares social learning curves across classes of networks:

- directed tree-like networks with different degree distributions;
- directed versus undirected networks with comparable degree distributions;
- bilateral-link networks versus clustered networks with triangles;
- Bayesian agents versus agents with correlation neglect;
- sparse versus dense networks for information aggregation.

Interpretation: The comparisons are theoretical counterfactuals: hold some network statistic fixed and change the structure of information paths.

4.2 Theorem 1: more links in directed networks

Under the bounded hazard rate condition, if a directed degree distribution first-order stochastically dominates another, then social learning and welfare are higher.

Interpretation: In directed locally-tree-like networks, more direct and indirect links provide more independent opportunities to hear about the product.

4.3 Theorem 2: directed beats undirected

If the neighbor-degree distribution satisfies the paper's stochastic dominance condition, social learning and welfare are higher in the directed network than in the analogous undirected network.

Interpretation: Backward links are not free information. They can make neighbors' non-adoption worse news, thereby lowering the usefulness of those neighbors as information sources.

4.4 Theorem 3: clustering reduces learning

For any degree d, clustered triangular networks generate lower social learning and welfare than bilateral networks with the same number of links.

Interpretation: A single adoption by any neighbor is enough to inform the agent. Correlated neighbors are less valuable than independent neighbors because they tend to fail together.

4.5 Theorem 4: correlation neglect reduces learning

In networks with cliques, agents who ignore correlation among information sources learn less than fully Bayesian agents.

Interpretation: Misperceiving redundant signals as independent makes the absence of adoption look too informative, reducing inspection and adoption.

4.6 Limits and assumptions

- Agents observe only adoptions, not failed inspections or entry times.
- Inspection reveals quality perfectly.

- Low-quality products are not adopted in the baseline model.
- The key monotonicity results depend on the bounded hazard rate condition.
- The tractable ODE systems are justified as limits of large random networks, not exact finite-network solutions.

Interpretation: The model is designed to isolate learning dynamics. It abstracts from strategic pricing, repeated interactions, and adoption externalities to focus on the informational role of network structure.

5 Tables, figures, and original equation panels

The paper has no empirical tables. Its visual content consists of network diagrams, learning curves, and original equation panels. The panels below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and baseline adoption equations

FIGURE 1.—**Illustrative networks.** The **left** panel shows an Erdős–Rényi network; social learning is described by a one-dimensional ODE (13). The **middle** panel shows a random network with triangles; social learning is described by a two-dimensional ODE (18)–(19). The **right** panel shows a random network with homophily; social learning is described by a four-dimensional ODE (42).

Figure 1: illustrative random networks.

ı the symmetric equilibrium is governed by the ODE

$$\dot{x} = \Phi(x). \qquad (\cdot$$

·es the idea that Kata's decision takes into account Lili's decision, which tak· ıt Moritz's decision, and so on. The simplicity of the adoption curve is in sta· the cyclical behavior seen in traditional herding models when agents obser· mmediate predecessors (Celen and Kariv (2004)).

2.2. *General Networks*

turn to the analysis of general random networks $\mathcal{G} = (I, \Xi, \mu)$. We first stu· ›ption rate and their social learning curves, and show that adoption rises wi· ·mation under a bounded hazard rate assumption. We then close the mod· sh that it admits a unique equilibrium.

with some definitions. As in the examples of Section 2.1, we generally d· i's probability of adopting product H by x_i. Agent i may not know t· nd so we keep track of agents' adoption across realizations of G and si· $x_{i,G,\xi}$ be agent i's *realized adoption curve*, given (G, ξ) after taking expect· others' entry times t_j and cost draws κ_j. Taking expectation over (G, ξ_{-i}), l· $_{,\xi_{-i}} \mu(G, \xi_{-i}|\xi_i)x_{i,G,\xi}$ be i's *interim adoption curve* given her signal ξ_i. Throug· ›p the time subscript t for these and all other curves.

agents form beliefs over their neighbors' adoption decisions. Since a sing· y one of i's neighbors perfectly reveals high quality, she only keeps track · Specifically, let $y_{i,G,\xi}$ be the probability that *at least one* of i's neighbors adop· by time $t \le t_i$ in network G given signals ξ, and $y_{i,\xi_i} := \sum_{G,\xi_{-i}} \mu(G, \xi_{-i}|\xi_i)y_{i,G}$· ·ctation conditional on ξ_i.

for i's realized adoption curve $x_{i,G,\xi}$, consider two cases. If she sees one of h· ıdopt, she updates her belief to $\pi_i = 1$ and adopts blindly. Conversely, if s· option, she updates her belief to $\pi_i = \pi(y_{i,\xi_i}) \le \pi_0$ and inspects and adop· ection cost is below this cutoff, $\kappa_i \le c_{i,\xi_i} := \pi_i$. Analogous to equation (2), · option curve follows:

$$\dot{x}_{i,G,\xi} = 1 - (1 - y_{i,G,\xi})\big(1 - F\big(\pi(y_{i,\xi_i})\big)\big) =: \phi(y_{i,G,\xi}, y_{i,\xi_i}). \qquad (\cdot$$

Baseline adoption-rate equations.

Read:

- Figure 1 shows the three main network structures studied: Erdos-Renyi, networks with triangles, and networks with homophily.
- The equation panel shows how adoption depends on social information y and the posterior $\pi(y)$ after no adoption.
- Social learning is summarized by the probability of seeing at least one neighbor adopt.

Economic message: The paper's key technical move is to reduce complex Bayesian updating to social learning curves that can be computed by ODEs.

5.2 Bounded hazard rate and directed-network ODEs

'he distribution of costs has a bounded hazard rate (*BHR*) *if*

$$\frac{f(\kappa)}{1-F(\kappa)} \leq \frac{1}{\kappa(1-\kappa)} \quad \textit{for } \kappa \in [0, \pi_0]. \tag{6}$$

ıs a bounded hazard rate, (6), *then i's interim adoption probability* x_{i,ξ_i} *mation* y_{i,ξ_i}.

ɩiating (5),

$$\text{›}(y, y) + \partial_2\phi(y, y) = 1 - F(\pi(y)) + (1-y)\cdot\pi'(y)\cdot f(\pi(y))$$
$$F(\pi(y)) - \pi(y)\cdot(1-\pi(y))\cdot f(\pi(y)), \tag{7}$$

ɩuality uses Bayes' rule (1) to show

$$-(1-y)\frac{\pi_0(1-\pi_0)}{\ldots} = -\frac{(1-y)\pi_0}{\ldots}\frac{1-\pi_0}{\ldots} = -\pi(y)\cdot(1-\pi(y)).$$

BHR condition and Lemma 1 derivative.

$$\dot{x}_\theta = E\left[\Phi\left(1 - \prod_{\theta'}(1-x_{\theta'})^{D_{\theta,\theta'}}\right)\right]. \tag{10}$$

ımensional ODE, which is easy to compute. Note that while the number grees is *infinite*, agents cannot observe their neighbors' degrees and so we ning problem at the level of neighbors' *finite* types. Thus, in the Twitter ;et one ODE each for celebrities, posters, and lurkers. In a regular, single-vith degree d, (10) simplifies to

$$\dot{x} = \Phi(1-(1-x)^d), \tag{11}$$

Directed-network ODEs.

Read:

- The bounded hazard rate condition makes adoption increasing in social information.
- The directed-network equation shows how a multi-type degree distribution maps into a system of ODEs.
- In a single-type regular network, the system collapses to $\dot{x} = \Phi(1-(1-x)^d)$.

Economic message: Directed locally-tree-like networks behave like independent information aggregators, so more links raise learning.

5.3 Figure 2: directed-tree social learning curves

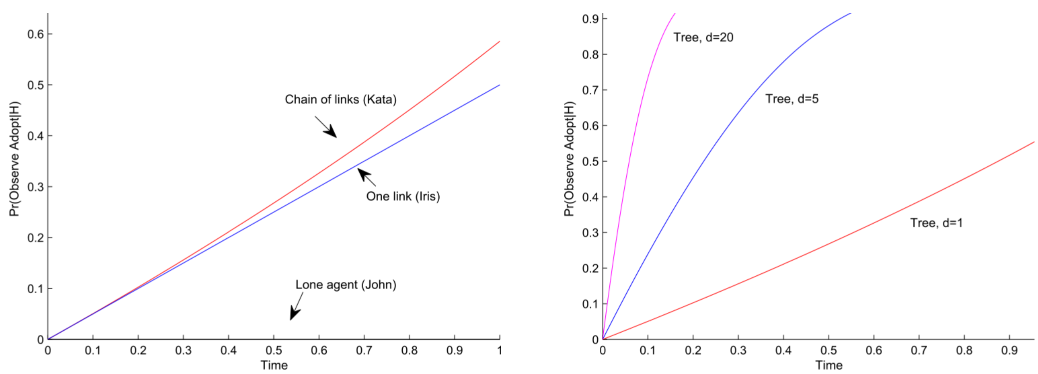

FIGURE 2.—**Social Learning Curves in Directed Tree Networks.** The **left** panel illustrates Examples 1 a 2. The **right** panel shows regular directed trees with degree d. This figure assumes uniform inspection co $\kappa \sim U[0,1]$, and an even prior, $\pi_0 = 1/2$.

Read:

- The left panel compares a lone agent, a one-link network, and a directed chain.
- The right panel shows regular directed trees with $d = 1, 5, 20$.
- Social learning curves rise more quickly as agents observe more direct and indirect links.

Economic message: In directed tree networks, connectedness accelerates diffusion because agents benefit from independent experimentation by direct and indirect neighbors.

5.4 Figures 3 and 4: backward links and undirected networks

FIGURE 3.—**Networks from Examples 3 and 6.** The **left** panel adds a backward link. The **right** panel a
orrelating link.

Figure 3: backward and correlating links.

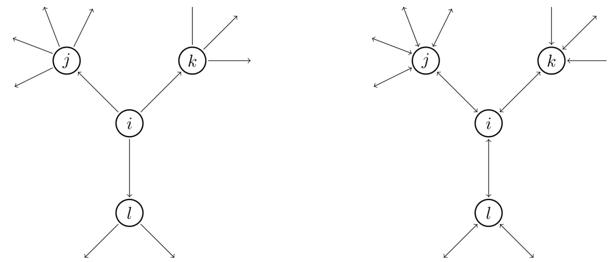

FIGURE 4.—**Comparing Directed and Undirected Networks.** This figure illustrates two Erdős-Rényi r
works from agent i's perspective, showing i's neighbors and their outlinks. For simplicity, the picture of
directed network does not show inlinks to i or her neighbors; in a large network, these inlinks do not aff
i's learning. Observe that i's neighbors have one more outlink in the undirected network, namely the link t
this reflects the friendship paradox.

Figure 4: directed versus undirected networks.

Read:

- Figure 3 illustrates two ways links can become harmful: backward links and correlating links.
- Figure 4 compares directed and undirected Erdos-Renyi neighborhoods.
- In the undirected network, i's neighbors have more outlinks, but those outlinks can include paths back to i.

Economic message: Not all links generate independent information. Backward links create self-reflection and can make neighbors' non-adoption more pessimistic.

5.5 Undirected and clustered ODE panels

$$\Pr\big(D' = d\big) := \frac{d}{E[D]} \Pr(D = d). \tag{12}$$

rdős–Rényi network, D is Poisson and $D' = D + 1$, whereas in a
$D \equiv d$.
ehavior of the limit economy as the number of agents grows large.
neighbors' adoptions as independent; Proposition 2′ in Section 3.5
. The simplest such network, corresponding to $D \equiv 1$, is the undi-
e 3. Following this example, we write $\bar{x}$ for the probability that i's
d at $t \le t_i$. With general degree distribution D, neighbor j addition-
er $D' - 1$ independent links, from which he observes no adoption
$)^{(D'-1)}$. All told, agent i expects j to observe an adoption with objec-
$- \bar{x})^{(D'-1)}$, while j expects to observe an adoption with the higher,
$1 - (1 - \bar{x})^{D'}$. So motivated, define $\bar{x}^*$ as the solution of

$$\dot{\bar{x}} = E\big[\phi\big(1 - (1 - \bar{x})^{D'-1}, 1 - (1 - \bar{x})^{D'}\big)\big] \tag{13}$$

Friendship paradox and undirected ODE.

it i's neighbors additionally learn from another $D' - 1$ independent triangles, from
hey observe no adoption with probability $(1 - \hat{x})^{2(D'-1)}$. All told, define $\hat{x}^*$ as the
n of

$$\dot{\hat{x}} = E\big[\phi\big(1 - (1 - \hat{x})^{2(D'-1)}, 1 - (1 - \hat{x})^{2D'}\big)\big]. \tag{15}$$

's actual, unconditional adoption rate then equals $E[\Phi(1 - (1 - \hat{x})^{2D})]$.

Clustered-network ODE.

Read:

- The undirected equation uses the neighbor-weighted degree distribution D'.
- The clustered equation tracks first adopters in triangles.
- Triangles raise correlation among neighbors' adoption decisions.

Economic message: The friendship paradox can partially offset the harm from undirected links, but clustering still reduces the value of information because sources become redundant.

5.6 Figure 5 and correlation neglect

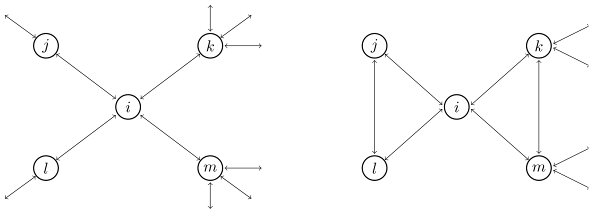

FIGURE 5.—**Bilateral Links vs. Triangles.** This figure illustrates two networks from agent i's perspective. I
the left network, everyone has $2D$ bilateral links, where $D = 2$ for $\{i, k, m\}$ and $D = 1$ for $\{j, l\}$. In the rigl

riangle formula (15),

$$\dot{\check{x}} = E\big[\phi\big(1-(1-\check{x})^{2(D'-1)}, 1-\big(1-\bar{x}^*\big)^{2D'}\big)\big]. \tag{17}$$

e first adopter in (i, j, k) expects to see an adoption with probability $\bar{y}^*_{2d} =$
', but the objective adoption probability is $\check{y}^*_{2d} = 1-(1-\check{x}^*)^{2(D'-1)}$.[17]
he effect of correlation neglect on social learning, consider an agent with $2d$
ork of triangles. Her social information is $\hat{y}^*_{2d} = 1-(1-\hat{x}^*)^{2d}$ in equilibrium
correlation neglect.

Figure 5: bilateral links versus triangles.

Correlation-neglect ODE.

Read:

- Figure 5 compares independent bilateral links with triangular cliques.
- Under correlation neglect, agents believe information sources are independent even when the true network is clustered.
- The resulting equation mixes subjective independent-link beliefs with objective triangle dynamics.

Economic message: Correlation neglect reduces learning because agents overreact to not seeing an adoption. They think the absence of adoption is more informative than it really is.

5.7 Figure 6: appendix Markov transitions

ork G and agents' signals ξ, and condition on a high-quality produc
then describe the distribution over states by $z_{\lambda,G,\xi}$ (as always omittir
on time t). Figure 6 illustrates the evolution of the state via a Markov ch
t example. Probability mass flows into state $\lambda = (\lambda_i, \lambda_j, \lambda_k) = (\emptyset, a, b)$
gent j enters and adopts, and from λ^{-k} as agent k enters and doesn't a

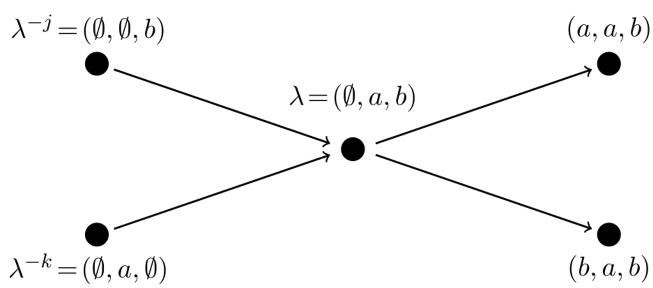

FIGURE 6.—Illustrative Markov Transitions underlying Proposition 1.

Read:

- The diagram shows the Markov transitions used to prove equilibrium existence and uniqueness.
- States record whether agents have not entered, entered and adopted, or entered and not adopted.
- Probability mass flows across states as agents enter and make adoption decisions.

Economic message: The appendix formalizes the ODE approach by showing how the high-dimensional finite-network process can be represented as a well-defined dynamical system.

6 Short summary

- The paper studies Bayesian learning and innovation diffusion in social networks.
- Agents observe only neighbors' adoption choices, not entry times or failed inspections.
- Inspection reveals quality perfectly, and adoption of the high-quality product is irreversible.
- The key state variable is a social learning curve: the probability of seeing at least one neighbor adopt.
- Under a bounded hazard rate condition, better social information increases adoption and welfare.
- Large directed random networks are tractable because local neighborhoods are tree-like and neighbors' signals are approximately independent.
- More links in directed networks increase learning and welfare.
- Undirected networks can reduce learning because of self-reflection through backward links.
- Clustering reduces learning because it correlates neighbors' decisions and makes information redundant.
- Correlation neglect reduces learning because agents treat correlated sources as independent and become too pessimistic after observing no adoption.
- Sparse networks can aggregate information well as average degree grows, but dense clustered networks may fail to aggregate information.
- The main methodological contribution is a low-dimensional ODE approach to Bayesian learning dynamics in large random networks.

7 Conclusion

7.1 Core conclusion

Board and Meyer-ter-Vehn show that social learning depends not only on the number of social links, but also on the structure of those links. Directed, locally-tree-like networks transmit information effectively. Undirected links, cliques, and correlated sources can reduce learning by creating self-reflection and redundant information.

7.2 Economic intuition

For an agent deciding whether to inspect an innovation, one adoption by any neighbor is enough to reveal high quality. Independent neighbors are therefore especially valuable. When neighbors' decisions are correlated, they tend to remain silent together, and this silence makes the agent more pessimistic. When agents ignore such correlation, they become even more pessimistic than a Bayesian would.

7.3 Takeaway

The paper turns a complex Bayesian network-learning problem into a tractable system of ODEs. Its central message is that network density is not always good: links help when they create independent sources of information, but can hurt when they create feedback, correlation, and redundancy.